Assignment: AI Workflow Assistant

Purpose

This assignment is designed to evaluate how you design, build, and explain a lightweight AI-powered workflow assistant within the next 2–4 hours.

The solution should demonstrate the ability to:

1. Accept a structured input request.
2. Process the request using an LLM.
3. Apply automated workflow routing logic.
4. Return structured output.
5. Document architecture decisions and tradeoffs.

Scenario

An internal team member submits this request:

"Create a customer success story for a retail customer adopting Adobe Experience Cloud."

The application should complete the following steps:

1. Intake

Accept the following structured input:

{

"title": "Customer Success Story",

"description": "Retail customer adopting Adobe Experience Cloud"

}

2. AI Processing

Use any LLM to generate:

- Summary
- Key messaging
- Recommended content format
- Complexity score: Low, Medium, or High

3. Workflow Automation

Based on the AI output, automatically route the request using code:

High complexity → Senior Review

Medium complexity → Content Team

Low complexity → Auto Approve

4. Structured Output

Return output in this format:

{s

"summary": "...",

"recommendedFormat": "...",

"complexity": "High",

"routingDecision": "Senior Review"

}

Technical Requirements

You may use one of the following approaches:

- Next.js or React
- Node.js
- TypeScript

Alternatively, you may build a pure Node.js solution with a simple frontend or CLI.

No database or authentication is required.

Keep the implementation intentionally lightweight; the goal is to evaluate technical judgment, AI integration, and workflow thinking, **not production-grade UI.**

Deliverables -

1. Working Application

The application may be one of the following:

- Single-page UI
- CLI tool
- Small web app

You may choose the format that best demonstrates their thinking.

A polished UI is not required.

2. Architecture Note

Submit a maximum 1-page architecture note explaining:

Architecture: Why this design was chosen.

AI Integration: How prompts were designed and how the LLM output is handled.

Automation Logic: Why the routing rules were chosen.

Scaling: What would change if the system handled 100, 1,000, or 10,000 requests per day.

3. Submission and Demo

Submit a GitHub link, ZIP file, or demo link before the interview, and be prepared to walk through the solution live during the interview.

Evaluation Criteria

- Technical judgment: The solution is simple, functional, and appropriate for the assignment scope.
- AI integration quality: Prompts, output handling, and assumptions are clearly explained.
- Automation logic: Routing decisions are implemented through code and are easy to understand.
- Scalability thinking: The architecture note identifies practical changes needed as request volume increases.